DataVault Mobile

Nigerian VTU & Bill Payment Business

Complete documentation for presentation,
understanding and running the business.
Flutter + Supabase + VTpass + Paystack

Demo works without keys. Live debits wallet via VTpass.

1. What is DataVault for?

- Mobile VTU shop: sell airtime, data and bills in Nigeria.
- Customer funds wallet, buys in 3-4 taps, gets token/PIN instantly.
- Owner earns margin + reseller storefronts.
- One Flutter codebase: Android, iOS, Windows, Web.

Demo (USE_EDGE=false) records success locally. Live (USE_EDGE=true) debits wallet atomically and fulfils via VTpass, funds via Paystack webhook.

2. Core features shipped

- Auth: onboarding, sign-up/in, password reset (Supabase Auth).
- Wallet + top-up. Live via Paystack init + webhook; demo credits directly.
- Catalog: airtime 4 networks, data Daily/Weekly/Monthly, electricity 11 Discos, cable DStv/GOtv/StarTimes, WAEC/NECO/JAMB/NABTEB, Smile/Spectranet/ntel, betting.
- Activity search+filters, receipt with token copy, beneficiaries, notifications, avatar, dark/light theme, 4-digit lock.
- Reference DVT-YYYYMMDD-XXXXXX, unique for idempotency.

3. Why customers choose us

3.1 1-tap Repeat + Buy again

- Home carousel of last 5 successes with Repeat amount button (44px).
- Repeat URL pre-fills provider/recipient/product/amount, jumps to Review.
- Also on receipt. Files: repeat_helper.dart, home.dart, purchase.dart.

3.2 Schedules & auto-buy

- Table schedules: kind/recipient/amount/frequency once/daily/weekly/monthly, next_run_at, active.
- Screen /app/schedules: create, pause, delete, Run due now. Auto-runs on Home start.
- Migration 002_schedules.sql.

3.3 Gift & Request via WhatsApp

- Gift tab shares receipt as wa.me text with token/PIN.
- Request tab builds need text + deep link for 1-tap fulfil.
- Buttons on purchase success + receipt. Files: gift_helper.dart, gift.dart.

3.4 Bulk dispatch

- Screen /app/bulk: airtime/data, network/plan, paste 100 numbers, valid/total counter, progress, batch ref.
- Each number charged once; failures skipped.

3.5 Storefront link

- Table storefronts: handle unique, display_name, markup_pct, active. Migration 003.
- Screen /app/storefront: handle, markup 0-20%, link https://datavault.ng/s/handle, WhatsApp share.

4. How money flows

- Funding: paystack-init returns authorization_url, browser checkout, webhook verifies HMAC + Paystack verify, inserts wallet_funding once, credits wallets.
- Purchase: vtu-pay auths user, debit_wallet locks row, inserts pending tx, calls VTpass /api/pay, code 000 = success + token else failed + refund.
- Bulk/schedules reuse vtu-pay. Requery via vtu-query.

- Never trust client; webhook + server verify are truth.

5. Tech stack

- Flutter 3, go_router, Manrope+Sora, Material 3, supabase_flutter, url_launcher.
- Supabase Auth, Postgres, Edge Functions Deno, Storage avatars.
- Edge: vtu-verify, vtu-pay, vtu-query, paystack-init, paystack-webhook.
- Secrets in Edge only; app keys via --dart-define.

6. Database

- profiles, wallets(balance_kobo), transactions(reference UNIQUE, status, metadata), beneficiaries, notifications.
- schedules(active,next_run_at), storefronts(handle unique).
- debit_wallet()/credit_wallet() atomic. RLS owner-only; storefronts public read when active.
- Run 001_wallet_ledger.sql, 002_schedules.sql, 003_storefronts.sql.

7. Setup & run

Demo: flutter pub get; flutter run -d chrome (windows needs Developer Mode).

Live: 1) run 3 SQL migrations. 2) supabase secrets set VTPASS_API_KEY, VTPASS_SECRET_KEY, VTPASS_BASE_URL, PAYSTACK_SECRET_KEY, APP_CALLBACK_URL, SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY. 3) supabase functions deploy vtu-verify vtu-pay vtu-query paystack-init paystack-webhook. 4) Paystack webhook to .../paystack-webhook. 5) flutter run --dart-define=SUPABASE_URL=... --dart-define=SUPABASE_ANON_KEY=... --dart-define=USE_EDGE=true

8. Running the business

- Pricing: price above VTpass cost; storefront markup 0-20%. Review weekly.
- Support: requery pending, failed auto-refunds, check webhook logs.
- Growth: referral bonus, 0.5-1% cashback, bulk for schools/churches, agents via storefront.
- Trust: token under 30s, WhatsApp receipt, always show reference.

9. Customer-friendliness audit

- Visibility: fixed dark-on-dark displayStyle bug, muted #6B7690 to #5B657D for AA 4.5. Measured muted/white 4.55, muted/bg fixed, primary/white 4.55, highlight 6.99, dark muted/bg 6.38.
- Consistency: cards radius 14, helpers cardColorOf/borderColorOf. Home migrated.
- Simplicity: 3-4 steps, 1-tap repeat, 44px targets. Next: Pidgin/local languages.
- Color: light AA pass; dark fixed for home/purchase/topup/bulk/schedules.

10. Costs & compliance

- VTpass cost + Paystack ~1.5% capped, Supabase/hosting, messaging.
- CAC, bank KYC, 5y logs, NDPR privacy, never store card PAN.
- Never expose service_role/VTpass/Paystack secret; reconcile daily.

11. Roadmap

- Full dark sweep, local languages, spend insights, family wallet.
- Reseller API, CSV upload, per-user virtual accounts, offline USSD fallback.
- Push, referrals, cashback engine, admin pricing dashboard.

12. File map

- lib/main.dart, app.dart, config.dart. lib/screens/: 15 screens including gift, bulk, schedules, storefront.
- lib/services/: vtu, paystack, schedule, storefront. repeat_helper, gift_helper, theme, models.
- supabase/functions x5, migrations x3, .env.example, README.md.